GUANLIN DEREK HUANG

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WEST LAFAYETTE, INDIANA

EDUCATION

Purdue University **Bachelor's Degree in Industrial Engineering**

Participating in a rigorous program involving courses such as but not limited to: Python for Data Analysis, Digital Systems Design, Advanced C Programming, Linear Algebra, Differential Equations, Workflow Analysis, Optimization, Stochastic Modelling

RESEARCH EXPERIENCE AND PROJECTS

Purdue Mechanism and Robotics Lab Undergraduate Research Assistant 10/2025- PRESENT

Dexterous Hand Manipulation Policy Development and Training | Purdue University (Prof. Yu She)

- Joining Principal Investigator Dr. Yu She of Purdue IE working towards novel solutions to robotic breakthroughs and policy trainings.
- Led the testing and development of a new robotic arm controller for both the UR5 and Panda Franka robotic arms, utilizing special simulations for testing such as mujoco, hardware development using CAD.
- Part of the dexterous hand manipulation team, focusing on VLA and VLM training policies.

Purdue Integrated Intelligent Electric Power Grid Lab Research Assistant 4/2025-10/2025

Midwest Power Grid Capacity Expansion Study | Purdue University (Prof. Andrew Liu)

- Conducted a research study surrounding the exploration of energy production in the Midcontinent region of the USA.
- Developed data inputs for a linear programming–based capacity expansion model (SWITCH) to study resource adequacy in the MISO region under decarbonization scenarios, incorporating stochastic modeling of renewable generation and load uncertainty**,** incorporating multiple weather-year scenarios
- Modeled generation portfolio evolution under fossil fuel retirements and renewable energy expansion through a linear optimization framework, incorporating capacity adequacy constraints and firm capacity requirements.
- Applied scenario-based and time-series analysis to evaluate impacts of renewable penetration, fossil retirements, emerging loads, and demand-side flexibility on system cost and reliability metrics alongside economic outlooks.

General Electric Robotics and Autonomous Systems Vertically Integrated Project 1/2024-5/2024

- Implemented forward and inverse kinematics for a 2-link robotic arm and developed the Jacobian matrix, applying Jacobian transpose methods to map Cartesian forces to joint torques.
- Designed a Cartesian-space PD controller for trajectory tracking along a nonlinear parametric path, generating, and following continuous task-space trajectories and converting them into joint-space commands.
- Built an iterative Jacobian-based inverse kinematics solver using pseudo-inverse methods with convergence and singularity handling.
- Simulated and validated robotic control pipelines in MuJoCo analyzing performance via trajectory visualization and mean squared error, and extending the system toward pick-and-place manipulation and motion planning

Machine Learning in Medical Imaging and Computational Biology 6/2021-8/2021

Semantic Segmentation and Classification of Liver Cancer Histopathology Images and Gene Mutation Prediction based on One-Shot Learning | MIT (Prof. Manolis Kellis)

- A research program led by MIT professor Manolis Kellis.
- My research was focused on using semantic deep learning to classify liver cancer biopsies.
- Grouped with two teammates to complete a research paper and presented research findings to Professor Kellis.
- Concepts applied in the program: Linear Algebra, Convolutional Neural Network, Machine Learning in Python, Data Analysis in R Programming.

Data Analysis in Real-Estate Market **5/2021-6/2021**

Beijing Rental Housing Market Study Focusing on Differing Metropolitan Areas | Tsinghua University (Prof. Mao Bo)

- Collected large-scale rental housing data from online real estate platforms using Python-based web polling techniques alongside data cleaning, preprocessing, and transformation in Python and Excel to handle missing values, inconsistencies, and outliers for ensuring structured and consistent datasets.
- Conducted exploratory data analysis to identify pricing trends, spatial distributions, and variability across districts, neighborhoods, and street-level granularity in Beijing. By applying statistical analysis and data aggregation methods to model rental price distributions and compare regional market dynamics.
- Developed data visualizations (charts, heatmaps, and distribution plots) to communicate key insights effectively.
- Produced a final presentation and presented to Professor Mao Bo of Tsinghua University

FIELD EXPERIENCE

Innogrit NAND Development and Validation Intern **6/2024-8/2024**

- Summer Intern at data storage technology company Innogrit
- Performed chip validation testing, identifying functional and performance issues during early development stages.
- Analyzed NAND characteristics such as endurance, latency, and error behavior to support design improvements.
- Conducted debugging and data analysis using I^2C and SMBus communication protocols.
- Generated detailed reports on test results and validation outcomes to assist in design iteration and optimization.
- Used a variety of tools during this internship such as Python, C (++), Verilog, Excel, Linux and more.

Purdue University Chinese Student and Scholar's Association **8/2025 - PRESENT**

- Took on the role of event planning secretary, focusing on logistics coordination for student events, including but not limited to scheduling, license acquisitions, promotion, and vendor communications.
- Doubled as vice chairman of the career development department, maintained alumni relationships through networking initiatives and outreach.
- Organized and hosted company information sessions providing students with opportunities to engage and learn from industry professionals.

ADDITIONAL

Languages: English (Native), Mandarin (Native)
Computer Literacy: Python, R, C, C++, Java, MATLAB, Verilog, Excel, LaTex